Grade: REVIEW REQUIRED

Assignment 1

Review Notes:
x Qa: Review manually.
x Qb: Review manually.
x Qc: Review manually.
x Qd: Review manually.
x Qe: Review manually.

1) a] $P[\geq 60] = P[60] + P[80] + P[100]$

$= .15 + .25 + .05$

$= .45$ or 45%

b.]

$\text{ExpectedValue} = \sum f(x)\,x$

$= 10(.10) + 20(.15) + 40(.30) + 60(.15) + 80(.25) + 100(.05)$

$= 1 + 3 + 12 + 9 + 20 + 5 = \boxed{50}$

$SD = \sqrt{\sum (x - \text{Mean})^2 f(x)}$

$= \sqrt{\sum (10-50)^2 .1 + (20-50)^2 .15 + (40-50)^2 .3 + (60-50)^2 .15 + (80-50)^2 .25 + (100-50)^2 .05}$

$= \cancel{\sqrt{160 + 2700 + 1200 + 900 + 18000 + 12500}}$

$= \sqrt{160 + 135 + 30 + 15 + 225 + 125}$

$= \sqrt{690} = \underline{26.27}$

c) It is the Mean assuming Normal Distribution.
It is not predictor of outcome of particular Withdrawl.

d) $P[\text{Mean} \pm 1SD] = P[50 \pm 26.27]$

$= P[23.73 \leq x \leq 76.27]$

$= P[40, 60] = \cancel{P[30]} = .30 + .15$
$= 45\%$

e) $P[\text{Mean} \pm 2SD] = \cancel{P[50]} +$

$P[50 \pm 52.54] = P[-2.54 \leq x \leq 102.54]$

$= P[10, 20, 40, 60, 80, 100] = \underline{100\%}$.